MedBrains Box: hardware reference design v0.1

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Summary

Build one small custom board, the MedBrains Box, for nodes and lab-analyser bridges, and buy the main server off the shelf for now. Memory and SSD prices in 2026 make a custom server dearer than a ready-made mini PC, while a node needs so little memory that a custom one costs about half.

- **Compute:** a Raspberry Pi Compute Module 5 (4 cores, 2 to 8 GB) plugged into our own carrier board.
- **What is custom:** a TPM chip, a hardware watchdog, power-cut hold-up, isolated serial ports for lab analysers, a sealed fanless case, a tamper switch and a recovery button.
- **Estimated parts cost per unit:** about $137 for a 2 GB node, $169 for a 4 GB node, $178 for a bridge, and $376 for a clinic server with one 256 GB SSD.
- **Status:** a design proposal. There is no schematic yet, and nothing here has been built or measured. Costs other than the module and SSD prices are my estimates, not quotes.

What MedBrains OS needs from hardware

Every requirement below comes from the MedBrains repo as of 10 November 2026: the README, `medbrains/os/README.md` and `RFC-HOSPITAL-OS`.

What the repo says	Hardware answer
The data disk is sealed to a TPM; a machine without one stops and asks	TPM 2.0 chip on the board
NVMe only for data, never an SD card	M.2 NVMe slots on the server build; nodes hold no patient data
The image is plain UEFI and a Pi 5 does not boot UEFI; a Pi image is planned (D2)	The board depends on that Pi image

What the repo says	Hardware answer
Active cooling is required on Pi-class boards	Metal case as the heatsink, with a fan header as fallback
Power loss mid-write, UPS with clean shutdown, hardware watchdog (§8o)	Supercapacitor hold-up and a watchdog on a small controller chip
Time is critical for clinical records	Battery-backed clock
Workstation: 2 cores, 4 GB; about 300 MB at the desktop plus 320 MB per open page	Node build with 2 or 4 GB
Small-clinic server: 4 cores, 8 GB, 64 GB SSD; the whole machine uses about 1.2 GB	Server build with 8 GB
Hospital server: 8 cores, 16 to 32 GB, mirrored SSDs, TPM, UPS, standby	Not this board: an off-the-shelf x86 server
On-site bridge listens for HL7 v2 from lab analysers and devices	Isolated RS-232 ports plus Ethernet
Real Wi-Fi cards need firmware the image does not carry	Wired first; the wireless module variant is optional

The repo also states that the OS has only run in virtual machines so far. This board would be its first real hardware.

Compute choice

The Raspberry Pi Compute Module 5 is the pick, because low-memory versions are the only cheap compute left in 2026 and MedBrains is small enough to use them. The 16 GB Pi 5 has gone from $120 at launch to $385, while the 2 GB module is $67.50.

Option	Processor and memory	Price, November 2026 (USD unless noted)	Verdict
Raspberry Pi CM5 Lite	4 Cortex-A76 cores; 2, 4, 8 or 16 GB	67.50, 100, 165, 375	Pick. Memory with error correction, in production until at least January 2036. Needs the Pi image, has no TPM and only one PCIe lane.

Option	Processor and memory	Price, November 2026 (USD unless noted)	Verdict
Radxa CM5	8 cores (RK3588S) with an AI chip; 4 to 32 GB	from €131 with 4 GB and 32 GB storage, VAT included, at one EU shop	Second source and AI upgrade. Fits most carriers of this style with caveats, and needs its own firmware work.
LattePanda Mu	4 Intel N100 cores; 8 or 16 GB	359, 499	Standard UEFI on x86, but too dear.
Ready-made N150 mini PC	4 Intel cores; 16 GB and a 512 GB SSD	281 to 349 complete	Best value for the server today. Not customisable.

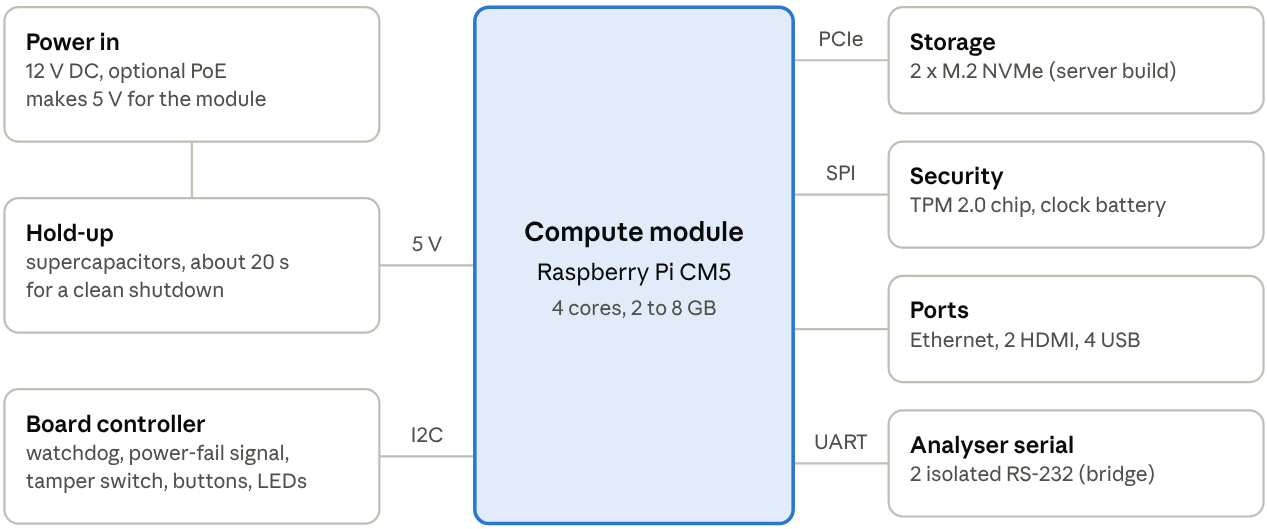
A custom server cannot beat the mini PC on price right now. The Pi module with 8 GB plus one 256 GB SSD already costs about $299 before any board or case, for half the memory and half the storage. The node is different: it needs no SSD and 2 to 4 GB, which no mini PC maker sells.

Designing the carrier around the Pi module connector keeps the Radxa and similar modules open as later options, to be verified pin by pin.

The carrier board

One circuit board serves all three builds; the node, bridge and server differ only in which parts are soldered on and which module is plugged in.

One board, three builds: only the fitted parts change



MedBrains Box carrier board · 7 subsystems around one compute module

The compute module is bought; everything around it is ours. The storage block is fitted only on the server build and the serial block only on the bridge build.

Subsystem	Example part	Why it is there
Power in	9 to 24 V input, 5 V 6 A converter, optional PoE module	One cable per node from a network switch, or a 12 V adapter that a cheap router UPS can back
Hold-up	2 × 50 F supercapacitors in series	About 19 s at 12 W or 29 s at 8 W, enough to stop the database cleanly; no battery to certify or replace
Board controller	RP2040 microcontroller, firmware in Rust	Restarts a frozen box, signals a power cut to the OS, logs case opening, drives the recovery button and status LEDs
Storage	PCIe switch (ASM1182e class) and 2 M.2 slots	Two drives on the module's single PCIe lane, for a mirror
Security	Infineon SLB9672 TPM, CR2032 clock cell	Disk key held in hardware; correct time after a power cut

Subsystem	Example part	Why it is there
Ports	1 Gb Ethernet, 2 HDMI, 2 USB 3, 2 USB 2	Screens, barcode scanners, label and receipt printers
Analyser serial	2 RS-232 ports, each behind a digital isolator and an isolated supply	Lab analysers are mains powered; isolation stops ground loops and protects both sides
Expansion header	I2C, SPI and USB on a front header	Later add-ons such as a staff-badge NFC reader or a small status screen

Parts and cost

A node comes to about $137 to $169 in parts and a clinic server to about $376, with the module and the SSD making up most of it.

Item (USD per unit)	Node, 2 GB	Node, 4 GB	Bridge, 4 GB	Clinic server, 8 GB
Compute module, list price	67.50	100	100	165
Carrier board, parts fitted (estimate)	39	39	39	39
Metal case and 12 V adapter (estimate)	21	21	21	21
microSD for the read-only system (estimate)	9	9	9	
Isolated serial ports (estimate)			9	
PCIe switch, hold-up, status screen (estimate)				17
256 GB NVMe SSD, contract price				134
Total, about	**137**	**169**	**178**	**376**

- A second SSD for a mirror adds about $134, taking the server to about $511.

- Module prices are the 3 November 2026 list. The SSD figure is the third-quarter 2026 contract price, up from about $33 a year earlier.
- Every line marked estimate is my figure for a run of about 100 boards. Expect it to move by 30% either way until a manufacturer quotes.
- Not included: GST, import duty, shipping, assembly labour, testing, certification, warranty and margin.
- Hold-up is fitted on the server build only, since nodes hold no patient data.

Enclosure, cooling and power

The case is a sealed aluminium box that doubles as the heatsink, so there is no fan to clog with dust and every face can be wiped with disinfectant.

- **Cooling:** the module presses against the lid through a thermal pad. The repo requires active cooling on Pi-class boards, so the fanless design stands only if a closed-case temperature test passes; a fan header stays on the board as the fallback.
- **Mounting:** behind a monitor for nodes, on a rail or wall for bridges, on a desk for the server.
- **Power budget:** 8 W for a node and 15 W for a server are my design assumptions, to be replaced by measurements.
- **Front:** power, health and network LEDs and a recessed recovery button. All cables leave from the back.
- **Tamper:** a switch under the lid tells the controller when the case is opened.

What changes in MedBrains OS

The board needs five pieces of software work, and the first one blocks everything else.

1. **Pi boot image.** The current image is plain UEFI and will not boot this module. The planned Pi image (D2 in `os/README.md`) is a prerequisite.
2. **TPM on arm64.** The sealed-disk path is tested only on x86-64. On this board the TPM sits on SPI, and as far as I know Pi firmware does not measure the boot into it, so the team must decide what the disk key is sealed to. The Pi's own signed-boot feature is the likely partner.
3. **Controller service.** A small Rust service talks to the board controller: watchdog heartbeat, shutdown on power cut, case-open events into the audit log, and temperature for the health screen.

4. **Read-only node storage.** Nodes boot from a microSD holding only the read-only system. That respects the rule against SD cards for data, but write volume on a node still needs measuring.
5. **Serial analysers.** The bridge listens for HL7 v2 over the network today. `FEATURES.md` lists ASTM protocol support as not done, so the serial ports have no software yet.

Certification in India

The box needs BIS registration before it can be sold, and anything used close to a patient needs a specialist's opinion first. I am not a compliance expert, so confirm each point with a BIS-recognised lab or consultant before the design is fixed.

- **BIS registration.** Computers fall under the Compulsory Registration Scheme as automatic data processing machines, IS 13252 (Part 1). Consultancies list servers, mini PCs and thin clients in that category, with sample testing at a BIS lab and a registration per brand.
- **Power adapter.** Adapters for IT equipment are registered separately under the same standard, so buy one that is already registered.
- **Radio.** The wired build has no Wi-Fi or Bluetooth. As far as I know, adding the wireless module brings a separate radio approval.
- **Patient vicinity.** A box at a reception desk or nurse station is IT equipment. One mounted at a bedside, or wired to a device that touches a patient, may fall under medical electrical safety rules (IEC 60601). Ask before promising bedside use.
- **Software claims.** The repo already notes that anything suggesting a diagnosis needs CDSCO approval. The hardware does not change that.

Risks and open questions

The biggest risk is that the OS has not yet booted on any real machine, so every hardware claim here is unproven.

- **Refurbished thin clients may be cheaper.** Used x86 thin clients already have UEFI and often a TPM. Price them locally before committing to a custom node; they are the baseline this board must beat.
- **Fanless cooling is unproven.** If the closed-case test fails, the case needs vents and a fan, which weakens the wipe-clean design.

- **One slow PCIe lane.** Two SSDs share a single 5 Gb/s lane. A clinic database should cope, but only a load test can say.
- **Prices keep moving.** Micron expects the memory shortage to last until at least 2028. Recheck every price before ordering.
- **Volume.** Design, tooling and registration are fixed costs. A custom board pays back over a few hundred units, not a few dozen.
- **Which customer first?** The RFC puts big hospitals first and says small clinics do not need the OS. Nodes serve big hospitals, many per site, so the node build fits that plan better than the clinic server does.

Plan

Prove the software on a bought development kit before spending anything on a custom board.

1. **Boot on a dev kit.** Buy one CM5 with Raspberry Pi's own IO board and get MedBrains OS running on it. This is the Pi image work and needs no custom hardware.
2. **Add the extras on the bench.** Wire a TPM module, an RP2040 board and an isolated serial adapter to the dev kit, and write the controller service.
3. **Measure and publish.** Memory, temperature in a closed metal case, power draw and hold-up time, added to the repo's measured tables.
4. **Design the board.** Schematic and layout in KiCad, reviewed by an experienced hardware engineer, then a first run of 5 to 10 boards.
5. **Pilot.** Run nodes at one friendly hospital alongside an off-the-shelf server.
6. **Register and sell.** BIS registration, then a production run.

Steps 1 to 3 cost little and answer most of the open questions above.

Sources

- MedBrains repository at commit `b35a16f` (10 November 2026): `README.md`, `FEATURES.md`, `medbrains/os/README.md`, `medbrains/RFCs/RFC-HOSPITAL-OS.md`
- PCWorld: Raspberry Pi 5 and Compute Module 5 prices, 3 November 2026
- CNX Software: Raspberry Pi 16 GB price rise and the memory shortage outlook
- element14: Compute Module 5 specifications
- DFRobot: LattePanda Mu prices

- Neven7: Radxa CM5 prices
- ComputingForGeeks: mini PC prices, November 2026
- Notebookcheck: SSD contract prices, third quarter 2026
- Bureau of Indian Standards: Compulsory Registration Scheme product list
- EVTL India: products registered as automatic data processing machines

Prices for the module and the SSD are from these pages. Several were read as search excerpts, not the full page, so check them before ordering.